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Capitalizing on Neighborhood Enterprise Zones: Are Detroit residents paying for the NEZ Homestead exemption?

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ABSTRACT

In this paper we examine the degree to which the tax benefits of the Neighborhood Enterprise Zone Homestead (NEZH) program are capitalized into the value of residential property. Although Enterprise Zone (EZ) programs and the studies examining these place-based policies are abundant, the NEZH program is unique because it offers tax abatements directly to the homeowner. Specifically, NEZH beneficiaries receive an 11.5 mill reduction (out of 66.61 total mills) for 12 years (excluding the value of land), and a phase-in to full taxation through year 15. We use residential sales data for properties sold in Detroit between 1997 and 2010 and a difference-in-differences model to estimate how much households are paying for the tax savings. Our primary results (intent-to-treat estimates) show all sales within NEZHs were 6–10% higher compared to those outside a zone. Furthermore, treatment-on-the-treated estimates highlight program beneficiaries paid 39% more for their property. With a full capitalization rate of 8.05%, these results indicate NEZH benefits are being overcapitalized by Detroit residents.

1. Introduction

Throughout the 1980s and 1990s it became common practice for state and local policymakers to incentivize the redevelopment of depressed cities and neighborhoods. These incentives were often in the form of 'place-based' policies that offered spatially targeted tax abatements, subsidies, public investments, and regulations. Kline and Moretti (2014) note spending on place-based policies has outpaced more traditional person-based assistance such as unemployment insurance. Enterprise Zone (EZ) programs quickly became one of the most popular place-based polices. By 2005, at least 43 states authorized EZ programs in one form or another (National Conference of State Legislatures, 2005). Most EZs were implemented to encourage businesses and individuals to remain, locate, or expand in depressed areas, raise property values, and expand the tax base (Engberg and Greenbaum, 1999). However, the specifics of each program vary substantially across states and local jurisdictions. For example, programs may provide income tax, property tax, or sales tax benefits, as well as a combination of all three.

Research examining Enterprise Zones has predominately focused on affected businesses. Topics have included job creation and retention (Peters and Fisher, 2002), business location decisions (Billings, 2009), and changes to commercial and industrial property values (Erickson and Syms, 1986; Landers, 2006).[1] The results have been mixed with research supporting the programs (e.g. Oakley and Tsao, 2006; Busso and Kline, 2008; Ham et al., 2011) and others questioning their effectiveness (e.g. Papke, 1994; Greenbaum and Engberg, 2004; Neumark and Kolko, 2010). Breaking the norm, Engberg and Greenbaum (1999) discuss the indirect effect of EZs on residential property values, again with mixed conclusions. Increased demand, improved infrastructure, greater employment, and higher earnings may generate higher prices within and around zones, but this may be offset if local taxes increase to pay for zone incentives.[2] We expand this literature by exploiting the unique nature of Detroit's Neighborhood Enterprise Zone program: a place-based policy targeted exclusively at residents rather than firms.[3] Specifically, we empirically examine

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[1] Ham et al. (2011) and Neumark and Simpson (2015) review the literature and highlight that the vast majority of place-based polices are targeted towards firms.

[2] Jurisdictions in other states may be interested in increasing property tax rates to pay for zone incentives; however, Michigan law prohibits this from happening (unless approved by voters). In addition, the City of Detroit has reached its maximum allowable rate.

[3] NEZs are not the only spatially targeted incentive providing tax relief to homeowners. We are aware of at least one other program in Michigan: Renaissance Zones (RZ). While RZ benefits are extremely generous to homeowners within the zone, effectively eliminating all state and local taxes for up to 15 years, the majority of RZs target businesses rather than residents. The NEZ program is the only spatially targeted benefit we are aware of that provides property tax relief *exclusively* to residents.

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whether program benefits in the form of reduced property taxes are capitalized into the value of Detroit property.

In 2006 Detroit implemented the first phase the Neighborhood Enterprise Zone Homestead (NEZH) program by creating 26 zones to encourage individuals to relocate to, or avoid leaving, the city. The second phase created 26 additional zones one year later. To qualify for the program, individuals must locate in a zone, apply for the program, and undertake a minimum of $500 worth of qualifying improvements within the first three years of being issued a NEZH certificate. NEZH beneficiaries receive a total reduction of 11.5 mills (out of 66.61 mills) in their property tax for 12 years and a phase-in to full taxation through year 15.[4] Assuming zero capitalization, this generates annual property tax savings of $330 for the average Detroit homeowner benefitting for the duration of the program, totaling nearly $4600 over the course of the program.[5]

To measure the degree NEZH benefits are capitalized in residential property values, we use sales data for properties sold in Detroit between 1997 and 2010. Our identification strategy uses a difference-in-differences (intent-to-treat) approach based on the geographic boundary of NEZHs, allowing us to compare properties within NEZH areas to non-zone properties, along with the timing of the NEZH implementation. We also estimate the dynamics of the capitalization rate and examine program take-up by considering differences between NEZH beneficiaries and non-beneficiaries (i.e. treatment effect on the treated). We mitigate potential bias from unaccounted neighborhood quality by using neighborhood fixed effects and we exclude the area immediately surrounding NEZH borders to eliminate potential spill-lovers across the boundaries. Finally, by examining properties within a single jurisdiction that are subject to the same statutory property tax rate we avoid spurious correlation between property taxes and un-observed public services prevalent in property tax capitalization literature (Gallagher et al., 2013).

Overall, our results show that (potential) NEZH benefits are being fully capitalized by Detroit residents. The difference-in-difference results indicate those within NEZHs pay approximately 6–10% more ($3400–$5500) for their property compared to those outside a zone and our dynamic estimates suggest these results are consistent over the first several years of the program. Taking program participation into account, treatment-on-treated estimates show significant overcapitalization by NEZH *beneficiaries* as they pay approximately 39% more for their property. These results are promising for financially struggling cities, such as Detroit. The estimated property value increase mitigates any short-term loss Detroit would have incurred if residents were undercapitalizing the tax break and leads to additional revenue during implementation and beyond zone expiration.

2. Neighborhood Enterprise Zone Homestead exemption[6]

Michigan legislature enacted the Neighborhood Enterprise Zone Act (PA 147 of 1992, amended) to provide distressed communities a place-based tool for encouraging individuals to rehabilitate, redevelop, and relocate to areas within their jurisdiction. Although PA 147 provides three different property tax reductions to residential property owners, Rehabilitation, New, and Homestead, this paper focuses solely on the Homestead program.[7]

To create a NEZH, a jurisdiction must qualify as an "Eligible Distressed Community" and be willing to provide property tax relief for the development and rehabilitation of residential housing.[8] Upon approval, the jurisdiction determines which areas will be designated as NEZs. The City of Detroit outlined, advertised, and began accepting applications for Phase I (26 zones throughout the city) of the NEZH program in 2005 and property tax relief for qualified residents started in 2006. One year later the city expanded the program to include 26 additional zones (Phase II). Fig. 1 shows the location of all 52 NEZH areas in Detroit. As seen in the figure, these zones cover a large portion of the city. In total, approximately 32,500 (or 28%) of the 113,000 improved, owner-occupied residential properties are located within a designated NEZH.

Residents must also qualify to participate in the program. The homeowner must locate within a zone after December 31, 1996, in a subdivision platted before January 1, 1968, and file an application for a NEZH certificate at the City Assessor's office. Furthermore, the property must be the applicant's principal residence and qualifying improvements of at least $500 must be made to the property within the first three years of being issued a NEZH certificate. Certificates may be revoked if the property is no longer the homeowner's principal residence, is noncompliant with local safety codes, or the owner is delinquent on their property tax payments. In addition, certificates are not transferable and new property owners must comply with the requirements to benefit from the program. Once qualified, NEZH beneficiaries receive a 50% reduction in *city operating* and *county operating* millage rates, excluding the land portion of their property, for up to 15 years. During the first 12 years of the program, eligible properties receive the full reduction, followed by a phase-in to full taxation with benefits reducing in equal intervals for the final three years.

Table 1 provides a comparison of millage rates for NEZH beneficiaries and non-beneficiaries. The total 2010 millage rate for owner-occupied residential property in Detroit was 66.61 mills. This indicates that homeowners pay $66.61 in property taxes for every $1000 in the taxable value of the property. The land portion of a NEZH beneficiary's property continued to be 66.61 mills, while the improved (non-land) portion was reduced to 53.82 mills. The reduction comes from a 50% decrease in the "General City Operating" and "Wayne County Operating" millages (as highlighted). The millage rates for the other taxing authorities remain the same as non-beneficiaries. Assuming land makes up ten percent of the total value of property in Detroit (the sample average), the overall millage rate for a beneficiary decreases to 55.1 mills.

Table 2 provides a different view of NEZH benefits by breaking down the tax payment of a $57,000 property in Detroit (the sample average), again highlighting differences between NEZH beneficiaries and non-beneficiaries. Continuing with the assumption that land is one-tenth the total value of property, the land and improved portions are worth $5700 and $51,300 respectively. The tax payment is calculated by multiplying the property's taxable value, equal to 50%

[4] One mill is defined as $1 per $1000 of taxable value.

[5] For the purposes of this paper, "homeowner" implies the property is the owner's principal residence. Being a principal resident is a requirement for the NEZH program.

[6] For additional details concerning Michigan's Neighborhood Enterprise Zone Program, see: http://www.michigan.gov/documents/taxes/NEZ_FAQ_276616_7.pdf.

[7] We do not examine capitalization effects from the Rehabilitation and New portions of the NEZ program for four reasons. First, these programs are very different from the Homestead program as they are directed toward the development of new/updated housing, rather than existing, and require substantially greater levels of investment (up to $7500 per unit). Second, substantial variations in program benefits exist between participants and the overall benefit levels are more generous compared to the Homestead exemption. For example, rehab beneficiaries get their property taxes frozen at the pre-rehabilitation amount (varies depending on the property) and one beneficiary had his taxable value frozen at $3026 for a condo he paid $300,000 for (2015 tax payment was equal $455). Third, the total benefit (net investment) is difficult to quantify since we do not know the level of investment each New and Rehab beneficiaries undertook, nor do we know the pre-rehab tax payments (important for determining benefits from the Rehab program). Finally, there is a lower number of beneficiaries, but more importantly, a much lower number of observations in our dataset. As of 2010, Detroit's Homestead program had more than twice the beneficiaries than the two other NEZ programs combined: approximately 7600 versus 3450. In addition, there were less than sixty "New" and ten "Rehab" properties that could be identified in our dataset. Inference based on only a few observations would be unreliable, especially given the wide variations in costs and benefits among New and Rehab program participants.

[8] For the requirements and a list of Michigan areas defined as "distressed," see: https://www.michigan.gov/documents/mshda/MSHDA-EDA-list_457982_7.pdf.

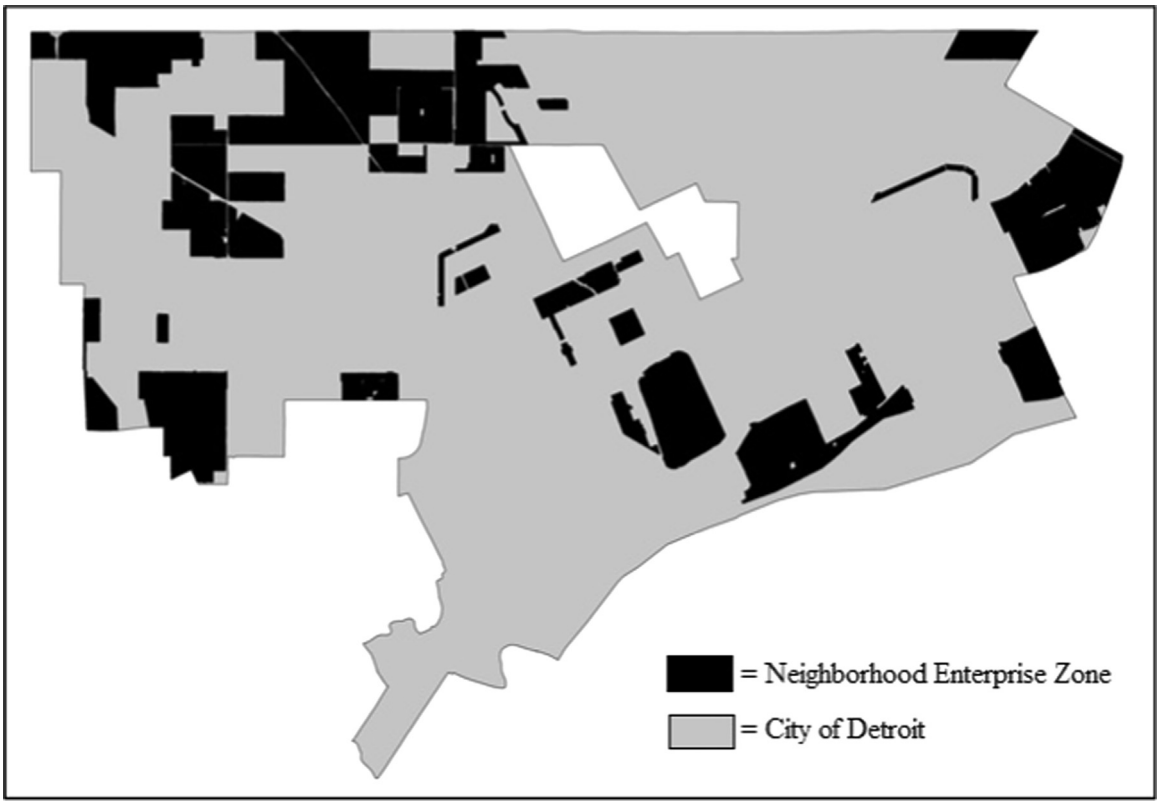


Fig. 1. Neighborhood Enterprise Zones in Detroit.

Table 1
Owner-occupied residential millage rates in Detroit (2010).

Taxing Authority	Non-Beneficiary Total	Beneficiary		
		Land	Improvement	Total*
State Education	6.00	6.00	6.00	6.00
General City Operating	**19.95**	**19.95**	**9.98**	**10.98**
Debt Service (City)	8.91	8.91	8.91	8.91
Library	4.63	4.63	4.63	4.63
School Bond Debt	13.00	13.00	13.00	13.00
School Operating	N/A**	N/A	N/A	N/A
School Judgment	0.10	0.10	0.10	0.10
Wayne County Operating	**5.64**	**5.64**	**2.82**	**3.10**
Wayne County	0.98	0.98	0.98	0.98
Wayne County Jail	0.93	0.93	0.93	0.93
Wayne County Parks	0.24	0.24	0.24	0.24
HCMA***	0.21	0.21	0.21	0.21
Wayne County RESA****	0.09	0.09	0.09	0.09
Wayne County RESA Sp. Ed.	3.36	3.36	3.36	3.36
Wayne County Comm. College	2.47	2.47	2.47	2.47
Wayne County Zoo	0.10	0.10	0.10	0.10
	66.61	**66.61**	**53.82**	**55.10**

* This assumes land makes up ten percent of the total value of property (sample average).
** Owner-occupied residential property (i.e. principal residents) are not subject to the school operating millage rate as a result of Proposal A (1994).
*** HCMA - Huron Clinton Metropolitan Authority.
**** Intermediate School District.

of the properties market value or "usual selling price" in an open market (Michigan Department of Treasury, 2011), with the tax rate (millage rate from Table 1 divided by 1000). Without capitalization (i.e. market value stays constant), the property tax payment is $1898 for non-beneficiaries and $1570 for beneficiaries – a difference of nearly $330.

Table 3 shows how the full capitalization rate for a $57,000 property is calculated, where full capitalization is defined as the increase in housing price (i.e. market value) equal to the present value of the stream of tax savings. As seen in Table 3, the tax payments for non-beneficiaries and beneficiaries in year 0 are identical to those calculated in Table 2. However, the tax savings from participating in the NEZH program are now negative because we include the minimum $500 investment homeowners are required to make in order to qualify for the tax reduction. Beyond year 0, two assumptions are required to calculate the present value (PV) of yearly tax payments: inflation rate and interest rate.[9] As a result of Proposal A, a taxable value cap approved by Michigan voters in 1994, year-over-year increases to a property owner's taxable value are limited by the lesser of the rate of inflation or 5%.[10] Historically, the rate of inflation has restricted the growth and we may simply increase the taxable value by this amount for each year. The interest rate affects the present value of future tax payments. Assuming the inflation rate is 3% and the interest rate is 2%, the total present value of tax savings is $4587 and the full capitalization rate is 8.05%.[11]

3. Empirical model

We use a hedonic pricing model to determine the degree NEZH

[9] We also recognize the length of homeownership may affect the capitalization rate. That is, the homeowner may change their capitalization payment if they don't expect to live in their home for the duration of the program. However, this is unlikely given the average length homeownership in Detroit is approximately 22 years, longer than program benefits are granted (Hodge et al., 2015).
[10] See Feldman et al. (2003) for an extensive discussion of Proposal A.
[11] The full capitalization rate also depends on the level of improvements a homeowner undertakes to qualify for the program. The rate decreases to 7.17% if $1000 worth of improvements are made and 0% if approximately $5100 worth of improvements are made. Unfortunately, we do not have the average level of improvements beneficiaries have incurred and we use the minimum level of investment since there exist no incentives for property owners to do more than the minimum. Therefore, our full capitalization rate represents an upper-bound based on the level of improvements.

Table 2
Tax payment comparison for $57,000 property (sample average).

	Total Market Value	Market Value (MV)		Taxable Value (TV)		Rate (Millage/1000)		Tax Payment (TV*Rate)		Total
		Land	Improvement	Land	Improvement	Land	Improvement	Land	Improvement	Tax Payment
Non-Beneficiary	$57,000	$5700	$51,300	$2850	$25,650	0.06661	0.06661	$190	$1708	$1,898
Beneficiary	$57,000	$5700	$51,300	$2850	$25,650	0.06661	0.05382	$190	$1380	$1,570

Table 3
Full capitalization rate calculation for $57,000 property (sample average).

Year	Taxable Value	Non-beneficiary Tax Payment (PV)	Beneficiary Tax Payment (PV)	Tax Savings
0	$28,500	$1898	$1570	($172)*
1	$29,355	$1917	$1586	$331
2	$30,236	$1936	$1601	$334
3	$31,143	$1955	$1617	$338
4	$32,077	$1974	$1633	$341
5	$33,039	$1993	$1649	$344
6	$34,030	$2013	$1665	$348
7	$35,051	$2033	$1681	$351
8	$36,103	$2052	$1698	$355
9	$37,186	$2073	$1714	$358
10	$38,302	$2093	$1731	$362
11	$39,451	$2113	$1748	$365
12	$40,634	$2134	$1765	$369
13	$41,853	$2155	$1876	$279
14	$43,109	$2176	$1988	$188
15	$44,402	$2198	$2103	$95
			Total Savings	$4587
		Full Capitalization Rate		8.05%

* Tax savings in year 0 subtract the minimum level of required investment ($500).

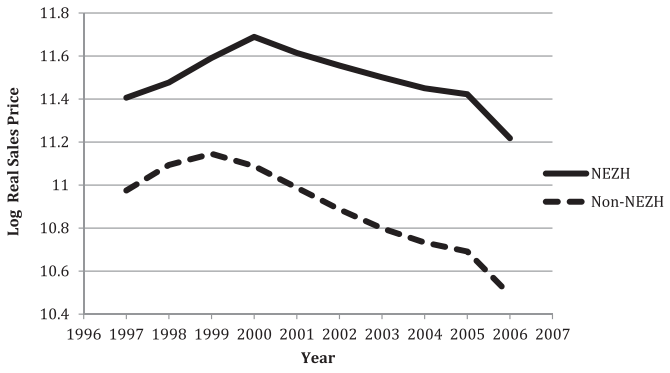


Fig. 2. Pre-NEZH sales trends. *Note:* includes the full sample of 24,355 observations.

benefits are capitalized into home values.[12] Our identification strategy builds off the quasi-experimental literature by utilizing a difference-in-differences approach,[13] which aids in overcoming issues with omitted variable bias that often plague traditional hedonic models. For example, the traditional hedonic model would be biased if NEZHs were located in more (or less) expensive housing areas due to unobserved factors (e.g. distance to amenities). Our analysis begins with the following difference-in-differences empirical model[14]:

$$\ln(\text{Price}_{ijt}) = \beta \text{postNEZH}_i + X_i\boldsymbol{\theta} + \text{Neighborhood}_j\Psi + YearMonth_t\boldsymbol{\phi} + \varepsilon_{ijt} \quad (1)$$

where Price_{ijt} is the real sales price of each house i in neighborhood j at time t,[15] X_i is a vector of the structural characteristics of house i (e.g. number of bedrooms, number of bathrooms, age of the house, etc.), Neighborhood_j is a neighborhood fixed effect, which comprise 490 neighborhood indicator variables provided by the City of Detroit,[16] and YearMonth_t is a vector of year-by-month dummy variables. Including neighborhood fixed effects controls for a wide range of observed and unobserved time-invariant spatial heterogeneity such as urban and natural amenities, crime rates across the city, among others. The neighborhood fixed effects also uniquely identifies the NEZH districts, such that the NEZH boundaries do not bisect neighborhoods. The time fixed effects control for temporal changes in the housing market such as seasonality and market trends, important because our sample overlaps the housing collapse during the Great Recession.

The objective of Eq. (1) is to provide a clear examination of whether those within NEZHs are capitalizing (potential) benefits into the price of their property. The variable of interest is postNEZH_i, a dummy variable indicating whether the property is in a NEZH after program implementation. The parameter β for the postNEZH_i variable represents the difference-in-differences estimate for the effect of the NEZH program on residential sales prices. It estimates the premium paid by those who bought a property in a NEZH and *could* benefit from the program compared to non-NEZH properties. Thus, the difference-in-differences model presented above is akin to an "intent-to-treat" estimate.

One of the key identifying assumption of the difference-in-difference model is that sales prices in NEZH and non-NEZH areas exhibit similar pre-existing trends (Angrist and Pischke, 2009). Fig. 2 displays sales price trends from 1997 through 2006. Although NEZH properties sold in this time period are more expensive than the non-NEZH properties (on average), the sales price trends are similar for the NEZH and non-NEZH properties prior to NEZH implementation. The only difference appears to be an increase in NEZH property values in the year 2000 without a corresponding non-NEZH sales price increase. The end of this time period shows the real-estate market collapse Detroit experienced at the beginning of the housing market crisis. The year-by-month fixed effects in our empirical strategy controls for the overall decline in the Detroit real estate market. Furthermore, we test the robustness of our results by estimating models using all of the available years of data (1997–2010), as well as leveraging the time period closer to NEZH implementation (2003–2010).

Choosing an appropriate control group is an important aspect of the difference-in-differences estimation strategy. Hanson and Rohlin (2013) argue that spatially-targeted redevelopment programs might induce spillovers across boundary lines. The authors find that areas bordering Federal Empowerment Zones experience a decline in business activity (number of establishments and employment) attributable to the policy. Thus, using areas prone to spillovers as a control group may bias results.[17] To mitigate potential spillovers across NEZH

[12] We follow the general theoretical approach outlined by Rosen (1974).
[13] See Parmeter and Pope (2013) for a detailed discussion on quasi-experimental techniques with hedonic property valuation.
[14] This specification represents the generalized difference-in-difference model. The time and neighborhood fixed effects take the place of the post-policy and NEZH district indicators in the traditional difference-in-difference model.
[15] Prices are adjusted to 2010 dollars using the Consumer Price Index (CPI).
[16] Our results are robust to alternative neighborhood fixed effects based on 961 census block groups in the City of Detroit.
[17] Although Hanson and Rohlin (2013) discuss negative spillovers, we recognize positive or negative spillovers could affect neighboring property. Positive spillovers may

boundaries, we also estimate models excluding homes that are geographically close to NEZH boundaries (within 0.15 miles on either side of a NEZH border). This strategy minimizes contamination of the control group from spatial spillovers of the NEZH program.

We also use the following specification to estimate the dynamic effect of the NEZH program on real sales prices:

$$\ln(\text{Price}_{ijt}) = \lambda_t \text{NEZH}_i * Year_t + X_i\boldsymbol{\theta} + \text{Neighborhood}_j\Psi + YearMonth_t\boldsymbol{\phi} + \varepsilon_{ijt} \quad (2)$$

λ_t is a set of coefficients for the interaction of the NEZH district variable and year dummy variables. Specifically, we interact the NEZH indicator with year dummies for 2004–2010, while using the same fixed effects strategy as Eq. (1) above. The coefficients of the interaction term show average differences between NEZH and non-NEZH properties relative to the base year 2003. This model highlights the dynamic response of homebuyers to the NEZH program by tracing out the difference between NEZH and non-NEZH properties over time. Furthermore, by including data prior to 2006 (program implementation year) we measure anticipatory effects of the program, along with a direct test of the difference-in-difference's pre-existing trends assumption.

4. Data and summary statistics

To conduct this analysis, we collected detailed information on sale prices and structural attributes of Detroit residential properties from a Michigan multiple listing service (MLS). All available data between 1997 and 2010 were collected, totaling 26,939 improved, residential sales.[18] Unfortunately, the MLS data did not provide neighborhood characteristics of each property, including whether or not the sale was within a NEZH, or the NEZH beneficiary status of a property. We geocoded and mapped all properties using a Geographic Information Systems (GIS) database and combined the properties with parcel-level NEZH maps provided by the city to identify the sales within NEZHs. Finally, the City of Detroit's Assessment division provided 2010 tax data identifying NEZH beneficiaries. Combining all information resulted in 2584 dropped observations since the addresses in the MLS dataset did not match those provided by the city. The final dataset includes 24,355 total sales and 20,764 sales excluding property within 0.15 miles of continuous NEZH boundaries.[19]

Summary statistics for the full sample and the sample excluding properties within 0.15 miles from continuous zone borders are provided in Table 4, along with variable definitions. There is little variation between the full sample and the non-border sample. Of particular interest, the average prices of both samples are approximately $57,000 and not statistically different at the 1% level. As noted earlier, NEZHs cover approximately 28% of the total improved, owner-occupied residential properties in the city. In comparison, our full and reduced samples show 32% and 29% of their sales in NEZH areas, respectively.

Finally, we assessed the comparability of the NEZH and non-NEZH samples prior to program implementation through a balancing test. Results of the balancing test can be found in Table A1 of the Appendix A. They show that none of the housing characteristics are statistically significant at the 10% level and *F*-tests of housing characteristics' joint significance yield p-values greater than 0.10 for the full and non-border sample. These results suggest strong statistical overlap for NEZH and non-NEZH property characteristics.

5. Results

5.1. Difference-in-difference results

Regression results are presented in Table 5. The first 2 columns use the full sample, while columns (3) and (4) use our subsample of properties that exclude sales within 0.15 miles of an NEZH border. In addition, columns (2) and (4) present the program effect using data from 2003–2010. One can think of these regression results within an intent-to-treat framework because our *postNEZH* variable pools beneficiaries and non-beneficiaries together. Again, tax records provided by the City of Detroit were utilized to determine the households that were NEZH beneficiaries by 2010. Nonetheless, residents purchasing homes within NEZHs could still become beneficiaries after 2010 (i.e. potentially intend to be treated).

In general, these results suggest that homebuyers in NEZHs pay a premium above those outside the districts. For example, the coefficient for *postNEZH* in column (1) suggests that homes within an NEZH sold for 6.38% (approximately $3400) above the baseline, after NEZH designation. This estimate is the premium paid after the policy came into effect, on top of these properties already selling for more than their non-NEZH counterpart prior to zone implementation (as depicted in Fig. 2). The premium increases to 9% when we consider a shorter time frame around NEZH implementation (column 2). Furthermore, columns (3) and (4) recognize the possibility of spillovers directly across NEZH boundaries and estimate the difference-in-difference model using the non-border subsample. The results are similar in magnitude and statistical significance, suggesting there were not significant positive or negative spillover over this geographic space.

The calculation presented in Table 3 shows that full capitalization of NEZH benefits for a $57,000 property would require an 8.05% premium ($4597). Our difference-in-difference results suggest that residents, on average, are fully capitalizing the potential tax savings from the NEZH program. We must keep in mind that these results include non-beneficiaries, where full capitalization of *potential* benefits may be interpreted as an overcapitalization of current benefits (where current benefits equal zero).

The full sample estimates from our dynamic model (Eq. (2)) are depicted in Fig. 3.[20] The NEZH interaction with indicator variables for the years 2004–2005 suggests there was not a strong anticipatory effect of the program. In addition, the dynamic results provide further credence to the credibility of the difference-in-differences design. The NEZH premium depicted by the coefficients in Fig. 3 grow to 10% right after program implementation. These estimates imply consistent overcapitalization of potential NEZH benefits based on the previously aforementioned full capitalization rate.

5.2. NEZH program take-up and treatment-on-treated effect

The difference-in-difference estimates provide an intent-to-treat estimate by pooling together those that do and do not benefit from the program. The coefficients measuring the intent-to-treat are comprised of two components: the NEZH take-up rate and the effect of actually benefitting from the NEZH program (i.e. the treatment-on-the-treated).

The take-up rate of the NEZH program was not universal among properties purchased within a qualified zone. Using data provided by the City of Detroit's Assessment Division, Table 6 shows take-up rates based on the year a property was purchased. The take-up rate was the

(*footnote continued*)
stem from NEZH properties being updated and fewer vacancies in NEZHs creating more attractive neighborhoods in the surrounding area. Negative spillovers may occur if a larger number of vacancies and less attractive neighborhoods are created in non-NEZH areas as residents leave for or are only attracted to more desirable NEZH neighborhoods – desirable due to the positive externalities discussed above as well as the lower tax rates.

[18] This dataset excludes sales with missing or unrealistic information (e.g. one property was excluded because the recorded lot size was 24,000 acres, one-third Detroit's total area), and extremely low values (i.e. sales less than $5000). Removing sales less than $5000 also helps to ensure that sales were arms-length transactions.

[19] Continuous borders are those not separated by natural boundaries (e.g. highways, rivers, railroads, etc.).

[20] The results for our dynamic model are robust when using the subsample (i.e. eliminating sales within 0.15 miles from an NEZH border).

Table 4
Summary statistics and variable definitions.

	Full sample		Non-Border Sample*		Definition
	Mean	Std. dev.	Mean	Std. dev.	
Price	57,642.18	45,522.71	57,839.22	45,277.55	Sale price of the residential property (2010 dollars)
NEZH	0.32	0.46	0.29	0.45	1=NEZH
postNEZH	0.13	0.33	0.11	0.31	1=NEZH properties purchased after NEZ designation
postBeneficiary	0.02	0.15	0.02	0.13	1=NEZH beneficiaries purchased after NEZ designation
Age	63.86	12.36	63.62	12.44	Age of the house
Living Area	1224.02	391.52	1205.47	344.11	Size of the house (square feet)
Lot Size	4934.72	1417.72	4896.67	1402.14	Size of the property (square feet)
Bedrooms	3.00	0.61	2.99	0.61	Number of bedrooms
Bathrooms	1.12	0.35	1.12	0.34	Number of full bathrooms
Basement	0.95	0.20	0.95	0.21	1=house has a basement
Garage	1.22	0.83	1.23	0.82	Number of cars the garage can hold
Stories	1.53	0.34	1.52	0.33	Number of stories
Split-Level	< 0.01	0.05	< 0.01	0.06	1=house is split-level
Sunroom	0.03	0.17	0.03	0.18	1=house has a sunroom
Masonry	0.61	0.49	0.60	0.49	1=house is full masonry
Part Masonry	0.05	0.23	0.05	0.22	1=house has partial masonry
Paid in Cash	0.49	0.50	0.48	0.50	1=house paid in cash
Conventional Mortgage	0.28	0.45	0.28	0.45	1=conventional mortgage
FHFA Mortgage	0.19	0.45	0.20	0.40	1=FHFA mortgage
# of Obs.	24,355		20,764		

* Excludes properties within 0.15 miles within an NEZH border.

Table 5
Difference-in-difference regression estimates.

	(1) All Sales	(2) All Sales	(3) Non-Border Sales	(4) Non-Border Sales
postNEZH	0.0638*** (0.0119)	0.0891*** (0.0143)	0.0695*** (0.0136)	0.0965*** (0.0164)
Property Characteristics	Yes	Yes	Yes	Yes
Neighborhood FE	Yes	Yes	Yes	Yes
Year-Month FE	Yes	Yes	Yes	Yes
Time Frame	1997–2010	2003–2010	1997–2010	2003–2010
Observations	24,355	14,870	20,764	12,331
R-squared	0.854	0.823	0.857	0.824

Notes: property characteristics include the variables listed in Table 3: age, size of living area, lot size, number of bedrooms, number of full bathrooms, basement, garage, number of stories, split-level, sunroom masonry as well as payment type.
Robust standard errors are in parentheses. Asterisks denote significance at the 1% (***), 5% (**), and 10% (*) levels.

highest for properties purchased in 2006 at 25%. The higher take-up rate immediately following program implementation matches expectations: properties purchased in 2006 had the longest time frame in our sample to undertake the requisite minimum of $500 in qualifying improvements. Table 6 also shows the average take-up rate for our sample is 16.2%. Dividing the point estimate of the NEZH premium by the take-up rate provides an estimate of the effect of the NEZH on program beneficiaries (i.e. treatment effect on the treated). For the full sample, this implies a premium of approximately of $23,000 (39.4%).[21] This is considerably higher than our full capitalization rate of 8.05%, and implies that beneficiaries are overcapitalizing NEZH benefits.

Furthermore, there is heterogeneity within NEZH districts on which properties become beneficiaries. Since the tax savings from the NEZH program come in the form of millage reductions, the benefits of the program depend on the assessed value of a property. Thus, higher

[21] This is calculated by using the implied average dollar amount from the NEZH premium (6.38% premium * $57,642 average real price) divided by the take-up rate of 16.2%.

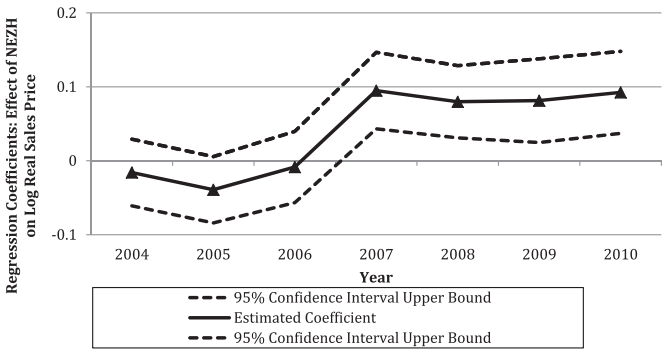


Fig. 3. Estimated dynamic effect of NEZH on real house prices (2003–2010).*Note:* Dashed line represents 95% confidence interval and solid line shows coefficient estimates for NEZH interacted with year variables.

Table 6
NEZH take-up rates by year property purchased.

Year	Take-Up Rate
2006	25.0%
2007	14.4%
2008	13.0%
2009	12.6%
2010	22.1%
Average Take-Up (2006-2010)	16.2%

Notes: The full sample of data consists of 3220 properties sold after NEZH program implementation, and 450 NEZH beneficiaries as of 2010.

valued homes have a greater incentive to become NEZH beneficiaries. Table 7 displays regression results from a linear probability model for take-up of NEZH beneficiaries among properties sold within qualifying zones. The regressions include year-month and NEZH district fixed effects to examine the differences in NEZH take-up for structural housing characteristics and the sales price. The results again match economic theory: higher valued properties are more likely to become NEZH beneficiaries. This result is statistically significant at the 1% level for both the full and non-border samples.

Table 7
NEZH program take-up estimates.

VARIABLES	(1) All Sales	(2) Non-Border Sales
Log Real Sales Price	0.0908***	0.0831***
	(0.0136)	(0.0162)
Age	0.000874	0.00206**
	(0.000762)	(0.000895)
Living Area	0.0312	0.0864***
	(0.0229)	(0.0296)
Lot Size	0.00831	0.00287
	(0.00547)	(0.00626)
Bedrooms	−0.0123	−0.0157
	(0.0130)	(0.0157)
Bathrooms	0.00468	−0.0181
	(0.0174)	(0.0214)
Basement	−0.0294	−0.0368
	(0.0337)	(0.0395)
Garage	0.00816	0.00197
	(0.00734)	(0.00898)
Stories	−0.0339	−0.0464*
	(0.0214)	(0.0254)
Split-Level	−0.0765*	−0.0816
	(0.0450)	(0.0529)
Sunroom	−0.0297	−0.0197
	(0.0378)	(0.0419)
Masonry	−0.00487	0.0108
	(0.0176)	(0.0197)
Part Masonry	−0.0145	0.00505
	(0.0223)	(0.0264)
Paid in Cash	−0.000379	−0.00277
	(0.0703)	(0.0847)
Conventional Mortgage	0.0599	0.0710
	(0.0718)	(0.0862)
FHFA Mortgage	0.0861	0.0880
	(0.0824)	(0.0989)
Constant	−0.498	0.0189
	(0.691)	(0.512)
Neighborhood FE	Yes	Yes
Year-Month FE	Yes	Yes
Observations	3220	2302
R-squared	0.150	0.174

Notes: regression results from a linear probability model where indicator for NEZH beneficiary status is the dependent variable.
Robust standard errors in parentheses
* p < 0.1
** p < 0.05,
*** p < 0.01,

6. Discussion and conclusion

In this paper, we exploit the unique nature of Detroit's NEZH program, a place-based policy targeted at residents rather than firms, to estimate whether benefits are capitalized into the value of residential property. Results show those within NEZHs pay 6 to 10% more for their property compared to those just outside a zone, and those benefitting from the program pay 39% more for their property. Comparing our estimates with the full capitalization rate of 8.05%, NEZH benefits are overcapitalized in Detroit property values and non-beneficiaries are capitalizing benefits not received: further evidence of overcapitalization.

These results should be encouraging to the struggling City of Detroit as they translate into additional property tax revenue. Excluding program administration and setup costs (e.g. advertising), the city receives approximately $3.7 million in additional tax revenue each year from the 7600 beneficiaries overcapitalizing the value of their NEZH benefits. From the remaining 24,900 properties within NEZHs (i.e. non-beneficiaries), the city receives approximately $3 million in additional annual revenue. These estimates are best-case scenarios as they assume Detroit residents, particularly NEZH homeowners, are paying their property taxes. As a worst-case scenario it has been estimated that 48% of Detroit residents are delinquent in their property taxes (Alm et al., 2014), reducing the total benefit of the city to approximately $3.2 million in additional revenue each year. Furthermore, Detroit may receive additional property tax revenue increases upon zone expiration as beneficiaries' millage rates are no longer reduced.

The estimated overcapitalization of residential property has several policy implications. Similar to Skidmore et al. (2012), our results suggest Detroit may lower tax rates to increase revenue through property taxes. This might be true in Detroit because the city's overall tax rate is particularly high relative to the surrounding communities in the region. Thus, place-based policies like the NEZH might provide a policy option to compete with other regions. Extending beyond Detroit, it may be possible for other financially strapped urban areas to leverage place-based policies as a potential revenue source. Our results also show a difference in the premium between beneficiaries and non-beneficiaries. This could suggest that some non-beneficiaries are experiencing a positive spillover from increased economic activity and qualified improvements made by the NEZH beneficiaries.

Nonetheless, it is not entirely clear why beneficiaries are over-capitalizing and non-beneficiaries are paying a premium; however, we currently have four hypotheses. First, increased competition for these houses may drive up the price. Halaas (2006) provides anecdotal evidence of potential increased competition and local awareness of the program in an interview at the beginning of the program: "The Neighborhood Enterprise Zone status was quite attractive to me. I looked at homes in Farmington Hills and Westland first, but when I checked out some places downtown and discovered how much I could save on property taxes because of the NEZ, my home search turned to finding a place in the city that I wanted to live." Related to increased demand, there may also be a decrease in demand for non-NEZH properties and the observed differential may partly result from reduced property values outside the zone (Merriman et al., 2008; Hanson and Rohlin, 2013). That is, it is possible these zones create a greater number of vacancies and less attractive neighborhoods in areas outside NEZHs as residents leave for or are only attracted to more desirable NEZH neighborhoods-desirable due to lower tax rates and positive externalities of improved property and (potentially) fewer vacancies. Second, implementing the NEZs may correspond with other city efforts to bolster these neighborhoods. Perhaps the city provides more, or better, public services to NEZ neighborhoods to keep the area thriving. We believe the results from our difference-in-difference models alleviate concerns that our estimates include other attributes being capitalized into property prices besides the NEZH property tax benefits. Third, non-beneficiaries may be benefitting from positive spillovers from already improved NEZ properties. Non-beneficiaries may also be anticipating participation in the program and capitalize the "potential" benefits they will receive once they are issued a NEZH certificate. In this case, the estimates for non-beneficiaries are reasonable as they are nearly identical to the full capitalization rate. Finally, NEZ homebuyers may be (over)capitalizing the one-year benefit they received from the previous homeowners' reduced tax payment, accumulated from Michigan's taxable value cap (Bradley, forthcoming).[22]

Although we find some positive effects from the NEZH program in Detroit, a number of caveats apply. First, the NEZH program in Detroit is different from EZ programs in other states. Therefore, it would be misguided to assume similar effects occur in other areas implementing

[22] Since the Detroit tax data and the MLS data do not provide the required information to repeat Bradley's study, we informally test this effect by including the length of the previous homeowner's tenure on the sale price of properties. To measure the length of previous homeowner's tenure, we restrict our data to repeat sales. There are 3010 multi-sale properties and 350 sales within 0.15 miles of continuous NEZH borders. Results indicate that property owners in Detroit do not capitalize previous homeowner's accumulated benefit. In addition, estimates measuring the capitalization of zone beneficiaries and those within zones from these regressions are statistically significant and larger than the results presented in Table 5.

place-based policies targeted toward businesses, especially in more affluent communities. Second, it is unknown what effect this program has had on vacancy and mobility rates, and thus the benefits may continue to be under- or overestimated. Especially with Michigan's taxable value cap policy, any reduced mobility created by the NEZH tax relief program may lead to lost revenue the city would have otherwise received (Hodge et al., 2015). However, this reduced revenue may be offset by reduced vacancy rates. Finally, finding positive effects of the program in Detroit does not immediately imply that it has been cost effective. This can only be fully answered when all costs are considered (e.g. advertising, enforcement, etc.). As state and local budgets get tighter the fiscal payback from place-based policies will likely come under greater scrutiny.

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Appendix A

See Appendix Table A1.

Table A1
Balancing test of pre-NEZH program structural housing characteristics.

VARIABLES	(1) All Sales	(2) Non-Border Sales
Age	4.52e−05	3.40e−05
	(4.60e−05)	(3.89e−05)
Living Area	0.00169	−9.93e−05
	(0.00206)	(0.00240)
Lot Size	0.000679	0.000391
	(0.000491)	(0.000462)
Bedrooms	0.000566	−0.000496
	(0.000707)	(0.000577)
Bathrooms	0.000172	−0.000622
	(0.00141)	(0.00112)
Basement	−0.000713	−0.000279
	(0.000621)	(0.000468)
Garage	0.000477	−0.000145
	(0.000389)	(0.000199)
Stories	0.00151	0.00147
	(0.00180)	(0.00165)
Split-Level	−0.000727	6.39e−05
	(0.000651)	(0.000740)
Sunroom	0.000896	0.000339
	(0.00195)	(0.00213)
Masonry	−0.000344	−0.000590
	(0.000758)	(0.000478)
Paid in Cash	0.00137	0.000961
	(0.00148)	(0.00165)
Conventional Mortgage	0.00227	0.00248
	(0.00167)	(0.00194)
FHFA Mortgage	0.000473	0.000920
	(0.00156)	(0.00182)
Constant	−0.0138***	−0.00554
	(0.00489)	(0.00792)
P-Value for F-test Housing Covariates	0.2961	0.8237
Neighborhood FE	Yes	Yes
Year-Month FE	Yes	Yes
Time Frame	1997–2004	1997–2004
Observations	11,847	10,450
R-squared	0.994	0.996

Notes: regression results from a linear probability model where indicator for NEZH district is the dependent variable.Robust standard errors are in parentheses. Asterisks denote significance at the 1% (***), 5% (**), and 10% (*) levels.

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